

Factors ameliorate pro-inflammatory microglia polarization through inhibition of reactive astrocytes induced by 2-chloroethanol

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ABSTRACT

Our previous studies have demonstrated that the crosstalk between astrocytes and microglia may trigger and amplify the neuroinflammatory response and, in turn, cause brain edema in 1,2-dichloroethane (1,2-DCE)-intoxicated mice. Moreover, findings from our in vitro studies showed that astrocytes are more sensitive to 2-chloroethanol (2-CE), an intermediate metabolite of 1,2-DCE, than microglia, and 2-CE-induced reactive astrocytes (RAs) can promote microglia polarization through releasing the pro-inflammatory mediators. Therefore, it is essential to explore therapeutic agents that may ameliorate microglia polarization through inhibition of 2-CE-induced RAs, which remains unclear till now. Results of this study revealed that exposure to 2-CE could induce RAs with pro-inflammatory effects, and fluorocitrate (FC), GIBH-130 (GI) and diacerein (Dia) pretreatment could all abolish the pro-inflammatory effects of 2-CE-induced RAs. FC and GI pretreatment might suppress 2-CE-induced RAs through inhibition of p38 mitogen-activated protein kinase (p38 MAPK)/activator protein-1 (AP-1) and nuclear factor-kappaB (NF-κB) signaling pathways, but Dia pretreatment might only inhibit p38 MAPK/NF-κB signaling pathway. FC, GI, and Dia pretreatment could all suppress the pro-inflammatory microglia polarization through inhibition of 2-CE-induced RAs. Meanwhile, GI and Dia pretreatment could also restored the anti-inflammatory microglia polarization via inhibition of 2-CE-induced RAs. However, FC pretreatment could not affect the anti-inflammatory polarization of microglia through inhibition of 2-CE-induced RAs. Taken together, findings from the present study demonstrated that FC, GI, and Dia might be the potential candidates with different characteristic for therapeutic use in 1,2-DCE poisoning.

1. Introduction

As an industrial organic solvent, 1,2-dichloroethane (1,2-DCE) is able to result in toxic encephalopathy under subacute poisoning and cerebral edema is the main pathological changes (Chen et al., 2015). Our previous studies have demonstrated that both astrocytes and microglia could be activated during the course of 1,2-DCE induced cerebral edema in mice, and the crosstalk between the activated astrocytes and microglia could trigger the neuroinflammation and eventually cause brain injury (Jin et al., 2019; Yang et al., 2021). Moreover, findings from our in vitro studies also disclosed that astrocytes were more sensitive to 2-chloroethanol (2-CE), an intermediate metabolite of 1,2-DCE, than microglia, and expression of interleukin-1beta (IL-1beta), tumor necrosis factor-alpha (TNF-alpha), and nitric oxide synthase (iNOS) were upregulated in 2-CE-induced reactive astrocytes (RAs), which could

stimulate microglia polarization through releasing these pro-inflammatory mediators (Wang et al., 2021). Therefore, it is necessary to uncover the agents that may ameliorate the microglial polarization through suppressing the RAs induced by 1,2-DCE or 2-CE.

Astrocytes can provide the trophic support for neurons and fulfil a range of functions in maintaining extracellular homeostasis and construction of blood-brain barrier (BBB) (Hawkins and Davis, 2005). Thus, they may be the first target for the chemicals from the blood into the brain. Studies up to now have demonstrated that astrocytes can be stimulated by various stimuli, and then experience the morphological, molecular, and functional remodeling to exhibit different gene phenotypes, called RAs (Escartin et al., 2021). Two kinds of RAs have been reported to be observed in neuroinflammation, which highly upregulated a serious of complement cascade genes and neurotrophic factors (Liddelow and Barres, 2015). The RAs with overexpression of classical

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complement cascade genes have been proposed to cause neurons destruction, while the others would be neuroprotective (Li et al., 2019; Liddelow and Barres, 2017).

As the resident immune cells in brain, microglial cells play the key roles in triggering neuroinflammation (Liddelow et al., 2017). They may be activated under pathological conditions, and secrete either pro-inflammatory factors enhancing cytotoxicity or anti-inflammatory factors benefiting tissue repair, depending on the type and magnitude of stimuli (Subhramanyam et al., 2019). Findings from our in vivo and in vitro studies proposed that the microglia might be activated by RAs, and the crosstalk between activated microglia and RAs could trigger and amplify the inflammatory response induced by 1,2-DCE in vivo or 2-CE in vitro (Wang et al., 2021; Yang et al., 2021). These results indicated that the neuroinflammatory response induced by 1,2-DCE or 2-CE could be ameliorated through inhibition of RAs.

Fluorocitrate (FC) is a well-established inhibitor of astrocytes, which is preferentially taken up by astrocytes and lead to reversible inhibition of tricarboxylic acid (TCA) cycle by targeting aconitase (Zhang et al., 2019). GIBH-130 (GI), the pyridazine-3-carboxamide derivative, is a novel pharmacological inhibitor of neuroinflammation, which exhibited the comparable efficacy of cognitive impairment relief to donepezil and memantine respectively in the transgenic murine AD models at a substantially lower dose (Zhou et al., 2016). Diacerein (Dia) is a non-steroidal anti-inflammatory drug that is reported to inhibit the signaling pathway associated with IL-1 β in the cells. Notably, Dia has recently shown to play the anti-inflammatory and anti-oxidant roles in multiple human diseases, and may have the potential therapeutic efficacy in these diseases (Almezgagi et al., 2020). Our previous in vivo studies have demonstrated that pretreatment with FC could markedly reverse the activation of astrocytes and microglia, as well as neuroinflammation in the brains of 1,2-DCE-intoxicated mice, suggesting that FC could abolish 1,2-DCE-induced microglial activation and neuroinflammation by effectively suppress the RAs (Yang et al., 2021). However, up to date, there is no report on the roles of either GI or Dia in the studies concerning 1,2-DCE-induced neuroinflammation.

Taken together, the above mentioned agents might play a therapeutic role in 1,2-DCE-induced neuroinflammation. Therefore, the objectives of this study were to explore the protective effects of FC, GI, and Dia on 2-CE-induced RAs, and their roles in microglial polarization by using different astrocytes and microglia co-culture systems.

2. Materials and methods

2.1. Antibodies and other reagents

Rabbit antibodies for p-p38 mitogen-activated protein kinase (p-p38 MAPK), p38 MAPK, p-p65, p-Inhibitor of kappaB α (p-Ik β) and p-c-Jun were obtained from Cell Signaling Technology, USA. FC was obtained from Sigma, USA. GI and Dia were the products of MedChemExpress, USA. ELISA assay kits for IL-1 β and TNF- α and Griess reaction for nitric oxide (NO) were purchased from Elabscience Biotechnology, China. The other antibodies and general reagents used in this study were obtained and prepared as our previously published paper described (Wang et al., 2021).

2.2. Primary astrocytes and microglia cultures

Rat astrocytes and microglia primary cultures were performed according to previously described methods (Wang et al., 2021).

2.3. Treatment of astrocytes

In the treatment of astrocytes, the cells were distributed in a control group, the inhibitor control group, a 2-CE group, and the intervention group (pretreated with FC, GI, or Dia), respectively. Astrocytes in the 2-CE group were exposed to 2-CE (30 mM) for 8 h. This time point was

chosen based on the results of the previous study (Wang et al., 2021). For the FC, GI, or Dia intervention, the pretreatment of astrocytes using FC (2, 1, or 0.5 μ M), GI (10, 5, or 1 μ M), or Dia (50, 30, or 10 μ M) for 1 h was carried out following 2-CE treatment (30 mM) for 8 h. The dose and time point of FC, GI, or Dia intervention were performed according to the results of the preliminary experiment and methods described previously (Bharti et al., 2017; Zhang et al., 2020; Zhou et al., 2016). Untreated astrocytes were used as control. FC (2 μ M), GI (10 μ M), or Dia (50 μ M)-treated astrocytes were used as inhibitor controls. Astrocytes were harvested at the end of 2-CE treatment and used in subsequent experiments. All these inhibitors were dissolved in dimethyl sulfoxide (DMSO), and the final DMSO concentration was less than 0.1%. The median lethal dose (LD₅₀) of 2-CE by oral was 71.3 mg/kg (Lawrence et al., 1971). The maximum tolerance dose of GI by oral was more than 2000 mg/kg (Zhou et al., 2016). According to the safety data sheet (SDS) provided by the manufacturer, the lowest toxic dose of Dia and the LD₅₀ of FC after oral administration were 10 mg/kg and 5.1 mg/kg, respectively.

2.4. Preparation of astrocyte-conditioned medium (ACM) co-culture system

The primary cultured rat microglia were co-cultured with ACM from 2-CE-untreated, 2-CE-treated, FC-pretreated + 2-CE-exposed, and GI-pretreated + 2-CE-exposed astrocytes, respectively, called control group, 2-CE treated group, FC intervention group, and GI intervention group. For 2-CE treated ACM preparation, astrocytes received the treatment of 2-CE (30 mM) for 8 h. For FC or GI intervened ACM preparation, astrocytes received the pretreatment of FC (2 μ M) or GI (10 μ M) for 1 h following 2-CE treatment (30 mM) for 8 h. ACM from 2-CE-untreated astrocytes was used as the control ACM. After the treatments, astrocytes were further washed with D-Hank's (DH) solution, followed by a 6-hour incubation in fresh media. Subsequently, these media were gathered and filtered by filters with the pore size of 0.22 μ m. Then, control ACM, 2-CE treated ACM, and FC or GI intervened ACM from astrocytes were added to the corresponding plates cultured with primary microglial cells of rats and incubated for 12 h. The time point was chosen based on the results of the previous study (Wang et al., 2021). After incubation, the microglia were gathered for the following experiments.

2.5. Preparation of transwell co-culture system

The transwell co-culture model of astrocytes and highly aggressively proliferating immortalized (HAPI) microglial cell line was adopted, as our previously published paper described (Wang et al., 2021). In the co-culture system, HAPI microglial cells were co-cultured directly with 2-CE-untreated, 2-CE-treated, and Dia-pretreated + 2-CE-treated astrocytes, respectively, called control group, 2-CE-treated group, and Dia intervention group. The current study cultured the astrocytes and HAPI microglial cells separately before co-culture. HAPI microglial cells were cultured in a 6-well plate without 2-CE exposure. Rat astrocytes grew on an inserted dish in the transwell plate. In the 2-CE-treated group, astrocytes were treated with 2-CE (30 mM) for 8 h. In the Dia intervention group, astrocytes were pretreated with Dia (50 μ M, 1 h) following 2-CE treatment (30 mM, 8 h). Untreated astrocytes were used as controls. Astrocytes growing on the inserted dish in the transwell plate were exposed to 2-CE via culture media. The media containing 2-CE was changed at the end of treatment, and astrocytes were rinsed twice by DH solution to remove the polluted 2-CE in the cells. The inserts containing astrocytes were then transferred to the corresponding 6-well plate containing HAPI cells and co-cultured for twelve hours. Therefore, when the HAPI microglial cells were co-cultured with astrocytes, the co-culture system was free of 2-CE. This time point was chosen based on the results of the previous study (Wang et al., 2021). HAPI cells, after being co-cultured, were harvested for the following experiments.

2.6. Immunocytofluorescence (ICF)

The method described in the published paper was followed strictly in this study (Wang et al., 2021). For quantification, the fluorescent intensity was measured using ImageJ v1.8.0 software (NIH, MD, USA) and calculated in three random microscopic fields of each section. Four sections per group were chosen for analysis.

2.7. Western blot (WB)

The procedure of western blot analysis described in the published paper was followed strictly (Wang et al., 2021).

2.8. Quantitative polymerase chain reaction (qPCR)

The method of qPCR described in the published paper was followed strictly in this study (Wang et al., 2021).

2.9. Enzyme-linked immunosorbent assay (ELISA)

Specific assay kits for the ELISA tests were used, and the experimental procedure described in the published paper was followed strictly in this study (Wang et al., 2021).

2.10. Quantification of nitric oxide (NO)

The accumulated production of NO in the media was assessed based on the Griess reaction. The method described in the published paper was followed strictly in the current study (Wang et al., 2021).

2.11. Statistical analysis

Statistical analysis was conducted with the SPSS (version 20.0, IBM, Armonk, USA). Data were described with mean $\pm$ standard deviation (SD). In the sample with only two groups, two independent sample *t*-test was used to evaluate the group mean difference between the two groups. In the sample with more than two groups, the one-way analysis of variance (ANOVA) was used to evaluate the group mean difference among groups. Once an ANOVA has given a statistically significant result, the Student-Newman-Keuls (SNK) post hoc testing was used to evaluate which specific pairs of group mean were different. For each parametric test, we confirmed the normality and variance homogeneity of the data. In addition, four independent experiments based on primary cells with different origins were carried out. The results were considered significantly different at $p < 0.05$.

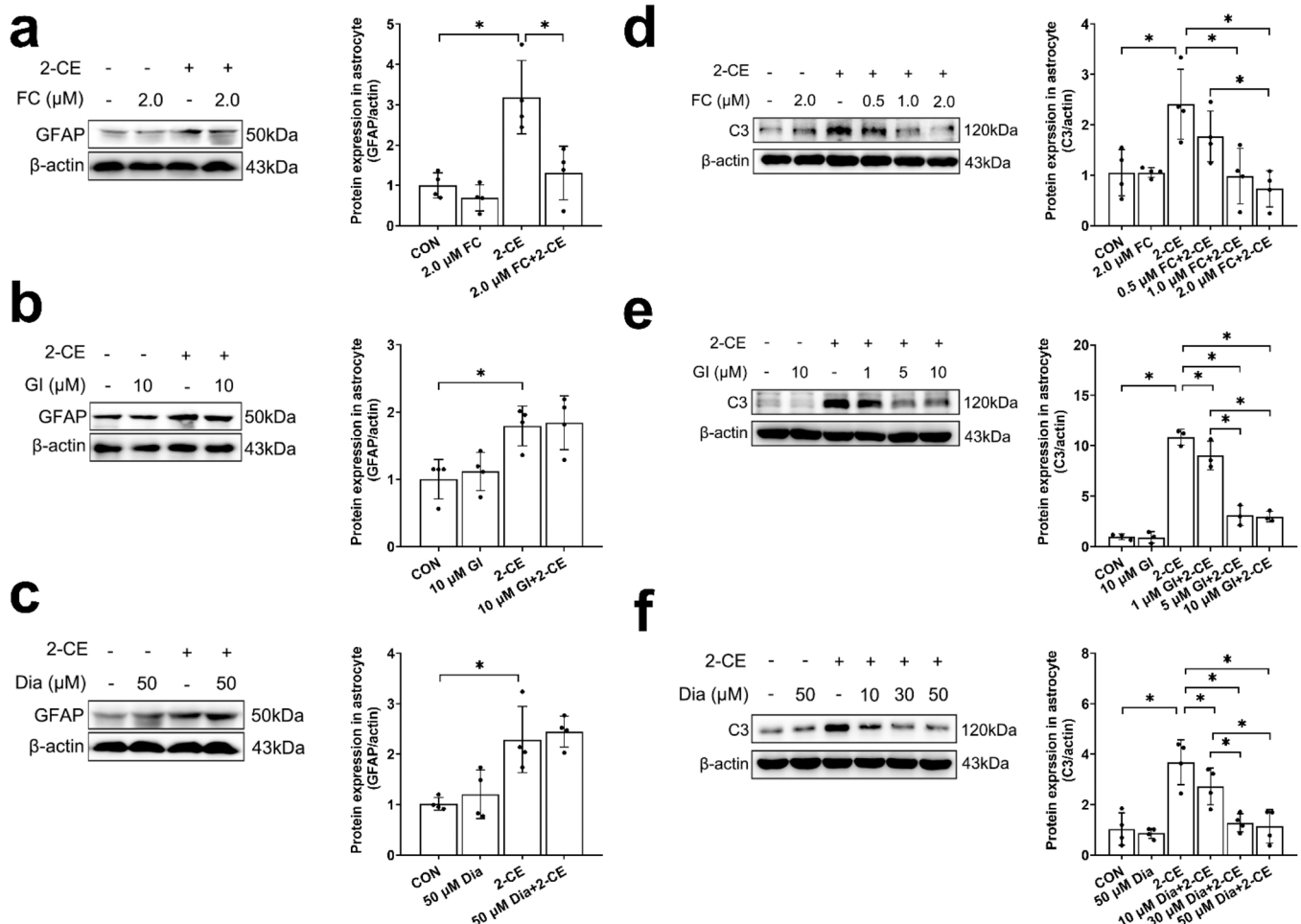


Fig. 1. Effects of pretreatment with FC, GI and Dia on protein expression of GFAP and C3 in 2-CE-exposed astrocytes. (a-c) GFAP protein levels in astrocytes were detected based on WB. The left panel showed representative blots. The right panel showed statistical analysis. (d-f) C3 protein levels in astrocytes were detected based on WB. ($n = 4$; mean $\pm$ SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, $*p < 0.05$). Notes: FC, fluorocitrate; GI, GIBH-130; Dia, Diacerein.

3. Results

3.1. Effects of pretreatment with FC, GI, and Dia on 2-CE-induced RAs

Glial fibrillary acidic protein (GFAP), the main protein component of the astrocytic intermediate filaments, is widely used to mark RAs due to its unique structure, and complement component 3 (C3) is the biomarker used commonly for identifying the RAs pro-inflammatory effects (Escartin et al., 2021; Liddelow and Barres, 2017). Similar to our previous results, the protein levels of both GFAP and C3 increased significantly in 2-CE-exposed astrocytes (Wang et al., 2021). As we observed, pretreatment with FC could significantly reduce the protein levels of GFAP. However, pretreatment with GI and Dia failed to affect its levels in 2-CE-exposed astrocytes. Besides, pretreatment with FC, GI, and Dia could all reduce the protein levels of C3 dose-dependently in 2-CE-exposed astrocytes (Fig. 1 and Fig. S1).

On the other hand, similar to our previous data, the protein levels of IL-1beta, iNOS, and TNF-alpha in 2-CE-exposed astrocytes, and levels of IL-1beta, TNF-alpha, and NO in their culture media increased significantly (Wang et al., 2021). Additionally, pretreatment with FC, GI, and Dia could all reduce markedly the levels of these pro-inflammatory mediators in either 2-CE-exposed astrocytes or their culture media (Fig. 2 and Fig. S2).

All these results clearly indicated that exposure to 2-CE could induce RAs with pro-inflammatory effects, and pretreatment with FC, GI, and Dia could all abolish the pro-inflammatory effects of 2-CE-induced RAs.

3.2. Effects of pretreatment with FC, GI, and Dia on p38 MAPK /activator protein-1 (AP-1) and nuclear factor-kappaB (NF-κB) signaling pathways in 2-CE-induced RAs

The p38 MAPK signaling pathway has been reported to regulate the production of pro-inflammatory cytokines through activation of transcription factors, such as AP-1 and NF-κB (Hoesel and Schmid, 2013; Ni et al., 2017). Our previous in vitro and in vivo studies have demonstrated that the p38 MAPK/AP-1 and NF-κB signaling pathways were activated in 2-CE-exposed rat astrocytes and in the brain of 1,2-DCE intoxicated mice during the course of brain edema formation (Jin et al., 2018; Jin et al., 2019; Wang et al., 2018). Furthermore, the p38 MAPK/AP-1 and NF-κB signaling pathways are closely related with the production and secretions of IL-1beta, TNF-alpha, and NO in 2-CE-exposed rat astrocytes (Wang et al., 2021). Therefore, the effects of pretreatment with FC, GI, and Dia on these signaling pathways in 2-CE-exposed astrocytes were explored in this experimental part.

Here we observed that the levels of p-p38, p-c-Jun, p-p65, and p-IκBα in 2-CE-exposed astrocytes increased significantly, which were similar to our previous results (Wang et al., 2018; Wang et al., 2017). Moreover, pretreatment with FC and GI significantly reduced the phosphorylation of p38, c-Jun, p65, and IκBα in 2-CE-exposed astrocytes. Pretreatment with Dia was capable to suppress the phosphorylation of p38, p65, and IκBα in 2-CE-exposed astrocytes. However, pretreatment with Dia could not affect the phosphorylation of c-Jun in these cells. Conversely, treatment with Dia alone could promote the phosphorylation of c-Jun in astrocytes (Fig. 3).

Findings from this part of experiment indicated that FC and GI pretreatment might suppress 2-CE-induced RAs through inhibition of p38 MAPK/AP-1 and NF-κB signaling pathways, and Dia pretreatment might

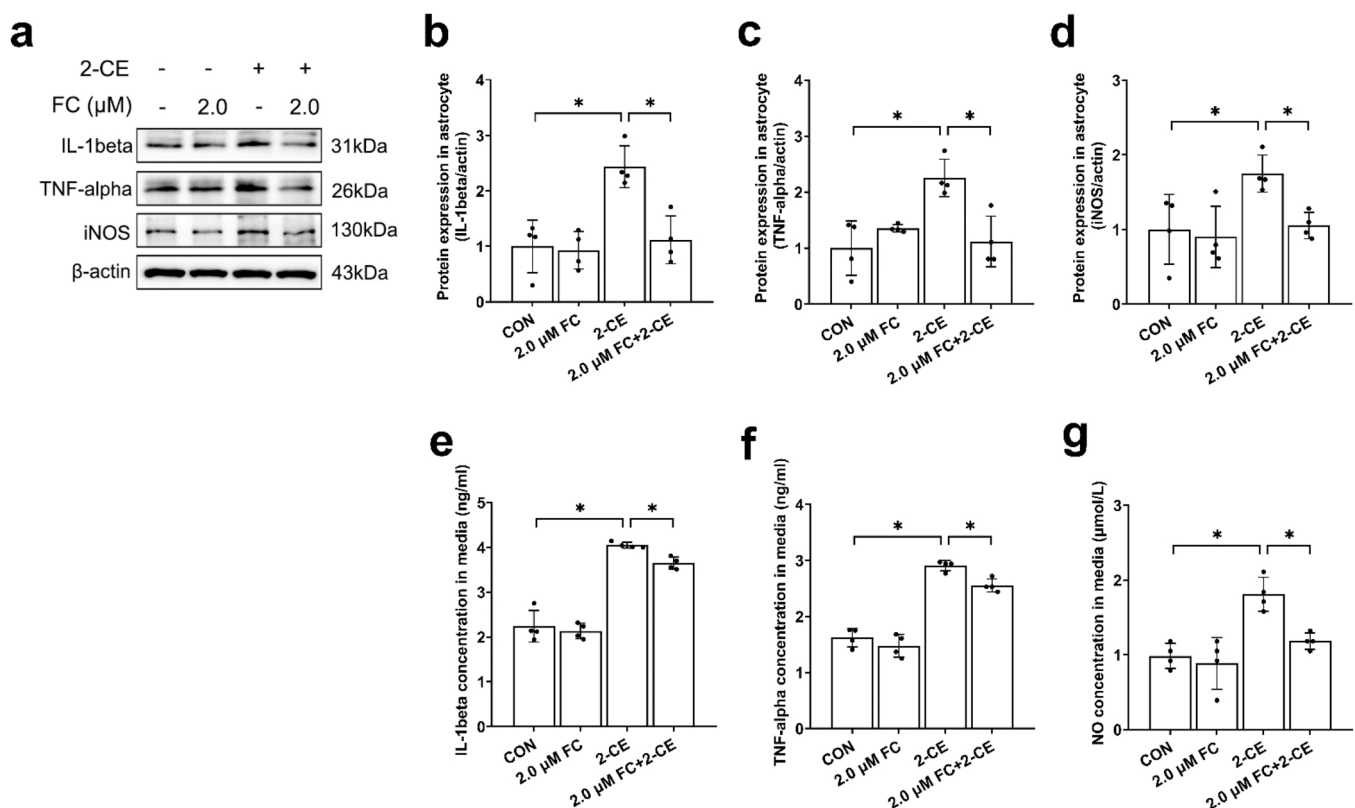


Fig. 2. Effects of pretreatment with FC on protein expression of IL-1beta, TNF-alpha, and iNOS in 2-CE-exposed astrocytes. (a-d) The protein levels of IL-1beta, TNF-alpha, as well as iNOS in astrocytes were detected via WB. The migrating bands at 31 or 26 kDa stand for the respective precursor of IL-1beta or TNF-alpha proteins. (e, f) ELISA was performed to quantify IL-1beta and TNF-alpha levels in culture media. (g) NO concentrations in culture media were measured by the Griess reaction. ($n = 4$; mean $\pm$ SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, * $p < 0.05$). Notes: IL-1beta, interleukin-1beta; TNF-alpha, tumor necrosis factor-alpha; iNOS, inducible nitric oxide synthase.

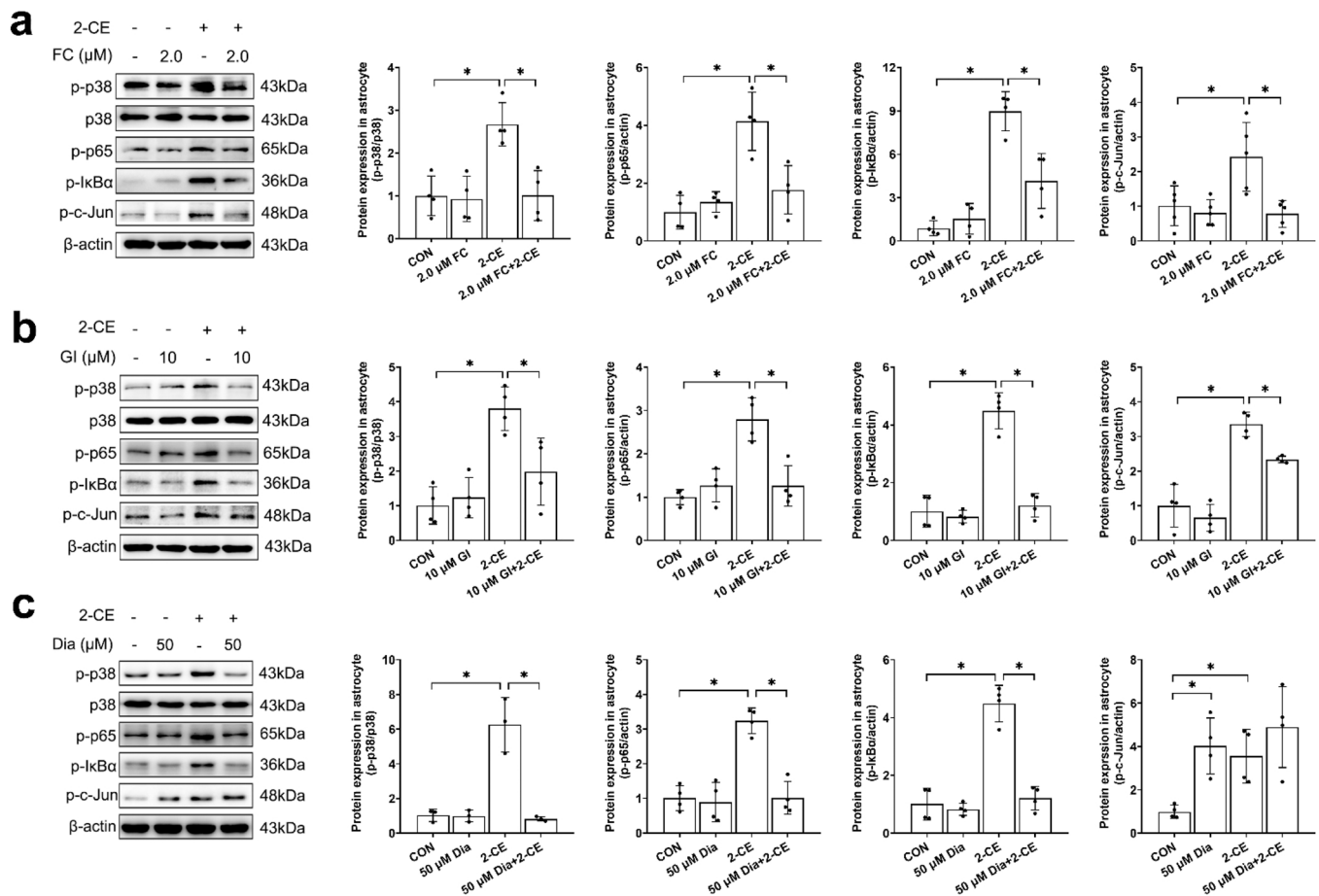


Fig. 3. Effects of pretreatment with FC, GI, and Dia on p38 MAPK/AP-1 and NF-κB signaling pathways in 2-CE-exposed astrocytes. (a) The phosphorylation of p38, p65, IκBα, and c-Jun in FC-pretreated astrocytes were detected via WB. (b) The phosphorylation of p38, p65, IκBα, and c-Jun in GI-pretreated astrocytes were detected via WB. (c) The phosphorylation of p38, p65, IκBα, and c-Jun in Dia-pretreated astrocytes were detected via WB. ($n = 4$; mean $\pm$ SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, $* p < 0.05$). Notes: p38 MAPK, p38 mitogen-activated protein kinase; NF-κB, nuclear factor-kappaB; AP-1, activator protein-1.

only inhibit p38 MAPK/NF-κB signaling pathway to suppress 2-CE-induced RAs. Since Dia treatment alone could activate the AP-1 signaling pathway in astrocytes, it might be not a suitable candidate for therapeutic use in 1,2-DCE poisoning.

3.3. Effects of pretreatment with FC, GI, and Dia on microglia polarization through inhibition of 2-CE induced RAs

With the aim of investigating the therapeutic agents for neuroinflammation induced by 1,2-DCE poisoning, in this experimental part, the effects of pretreatment with FC, GI, and Dia on microglia polarization through inhibition of 2-CE induced RAs were explored by using different co-culture systems with astrocytes and microglia. The cellular protein levels of cluster of differentiation 86 (CD86) and arginase-1 (Arg-1) are commonly detected to be the biomarker of pro-inflammatory and anti-inflammatory polarization of microglia, respectively (Li et al., 2018; Wu et al., 2018). The protein expression of ionized calcium binding adapter molecule 1 (Iba1) was detected to mark the microglia (Dhaya et al., 2021). At first, the primary cultured rat microglia were co-cultured with ACM from 2-CE-untreated, 2-CE-treated, FC-pretreated + 2-CE-exposed, and GI-pretreated + 2-CE-exposed astrocytes, respectively, called control group, 2-CE treated group, FC intervention group, and GI intervention group.

Here we found that the protein and mRNA levels of CD86 in rat microglia of 2-CE treated group increased significantly compared to control group. In contrast, those in either FC or GI intervention group

decreased significantly compared to 2-CE-treated group. Conversely, the protein and mRNA levels of Arg-1 in rat microglia of 2-CE-treated group decreased significantly compared to control group. However, only those in GI intervention group increased significantly compared to 2-CE-treated group (Fig. 4). The similar changes of CD86 and Arg-1 protein expression in rat microglia detected by co-ICF staining among groups were observed (Fig. 5).

It is time-consuming and technically challenging to obtain sufficient pure (>95%) cultures of rat microglia for experiment. To circumvent these challenges, HAPI microglial cells were used in this experimental part for investigating the effects of pretreatment with Dia on microglia polarization via inhibition of 2-CE-induced RAs by using transwell co-culture system. In this co-culture system, HAPI microglial cells were co-cultured directly with 2-CE-untreated, 2-CE-treated, and Dia-pretreated + 2-CE-treated astrocytes, respectively, called control group, 2-CE-treated group, and Dia intervention group. Here we observed that the protein and mRNA levels of CD86 in HAPI microglial cells of 2-CE treated group increased significantly compared to the control group. In contrast, those in Dia intervention group decreased significantly compared to 2-CE-treated group. Conversely, the protein and mRNA levels of Arg-1 in HAPI microglial cells of 2-CE-treated group decreased significantly compared to control group, whereas those in Dia intervention group increased significantly compared to 2-CE-treated group (Fig. 6).

Taken together, findings from this part of experiment disclosed that FC, GI, and Dia pretreatment could all suppress the pro-inflammatory

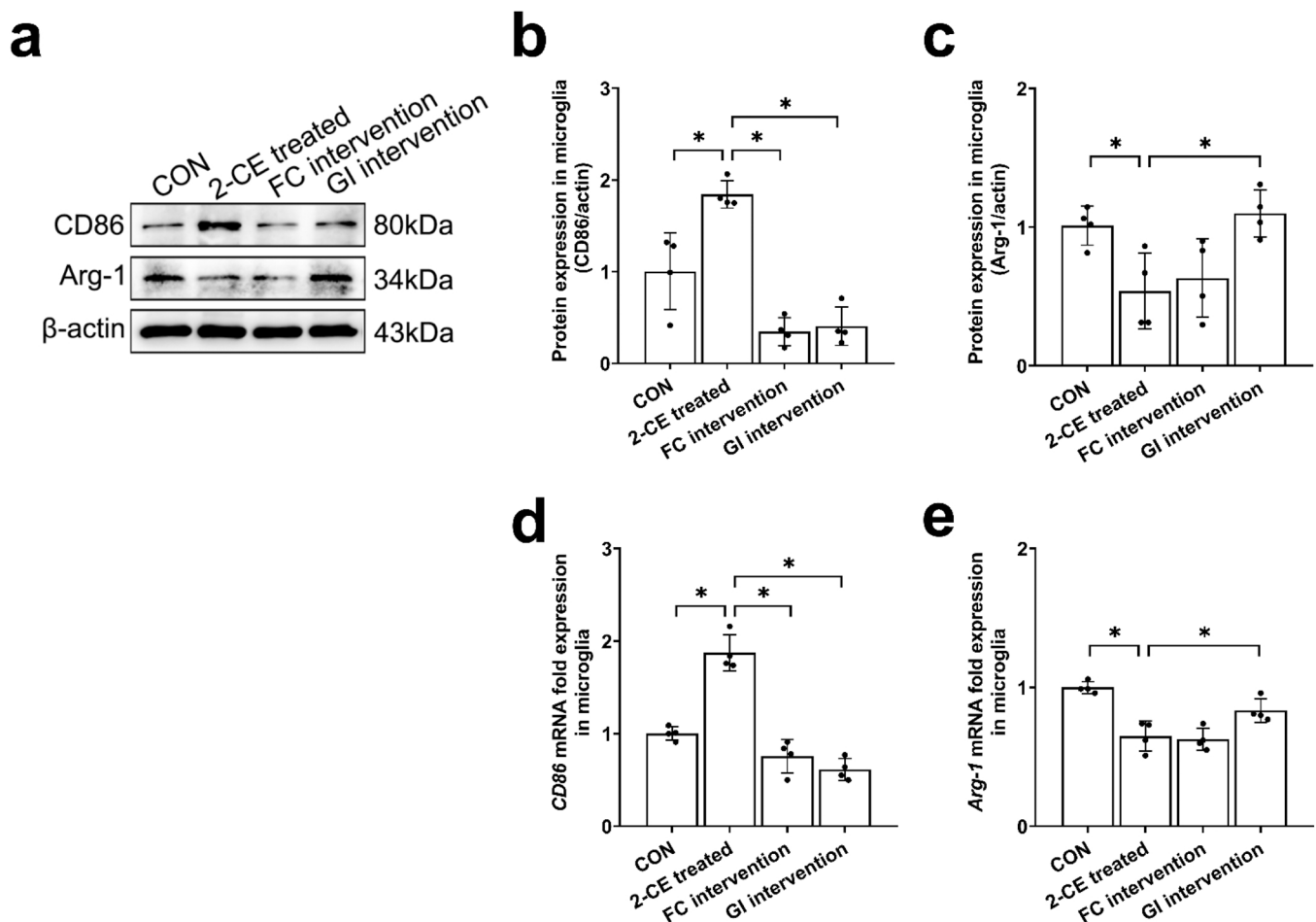


Fig. 4. Effects of pretreatment with FC and GI on microglia polarization via inhibition of 2-CE-induced RAs. (a) The protein levels of CD86 and Arg-1 in microglial cells of rats which were co-cultured with different ACM were detected based on WB. The panel showed representative blots. (b, c) The panel showed statistical analysis of the protein levels of CD86 and Arg-1 in microglial cells of rats. (d, e) qPCR was used to assay the *CD86* and *Arg-1* mRNA levels in rat microglia. ($n = 4$; mean $\pm$ SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, $* p < 0.05$). Notes: RAs, reactive astrocytes; ACM, astrocyte-conditioned medium; CD86, cluster of differentiation 86; Arg-1, arginase-1.

microglia polarization through inhibition of 2-CE-induced RAs. Meanwhile, GI and Dia pretreatment could also restore the anti-inflammatory microglia polarization via inhibition of 2-CE-induced RAs. However, FC pretreatment could not affect the anti-inflammatory polarization of microglia through inhibition of 2-CE-induced RAs.

4. Discussion

As recently reported, two types of RAs might play the pro-inflammatory or anti-inflammatory roles through upregulating different genes (Liddelow et al., 2017). Over the past decade, the complement system has emerged as a pivotal regulator of astrocytic functions, and C1q, C3, and C5 are the main complement proteins involved in the RAs (Pekny et al., 2019). The C3 has been reported to be expressed highly in the RAs with pro-inflammatory effects, but not expressed in the RAs with anti-inflammatory effects, thus the C3 protein expression can be used as a marker to identify the kinds of RAs (Li et al., 2019). As reported, the mice with C3 knockdown or pretreated with C3a receptor (C3aR) antagonist showed reduced infarct volume and less neurological impairment after ischemia (Mocco et al., 2006). Similarly, treatment with C3aR antagonist could improve the neurologic outcome by reducing the infiltration of inflammatory cells after experimental intracerebral hemorrhage (Rynkowski et al., 2009). These results clearly indicated that the C3 might play a very important role in triggering the pro-inflammatory effects during brain injury.

Accumulating evidence suggests that RAs with pro-inflammatory or anti-inflammatory effects may play the detrimental or protective functions at different stages of the pathological process (Escartin et al., 2021). Our previous data showed that treatment with 7.5 mM 2-CE could mainly induce RAs with neuroprotective functions, and 30 mM 2-CE could induce dominantly RAs with C3 overexpression (Wang et al., 2021). The present results revealed that treatment with 30 mM 2-CE induced RAs with the pro-inflammatory effects, which were similar with our previous findings (Wang et al., 2021). Additionally, the current study demonstrated for the first time that the supplementation of exogenous FC, GI, or Dia could all abolish the pro-inflammatory effects of 2-CE-induced RAs, and their effects on 2-CE-induced RAs could also affect the polarization of co-cultured microglia, suggesting that these agents might be the candidates to be therapeutic use in 1,2-DCE poisoning.

FC is the well-accepted agent for inhibiting the RAs and widely used for studying the roles of RAs under different pathological conditions (Shang et al., 2015; Wang et al., 2019). However, the role of FC in 2-CE-induced RAs has not been reported yet. Because FC can inhibit reversibly the TCA cycle in astrocytes by targeting aconitase, treatment with FC in fact, may reduce the cellular activity of RAs (Zhang et al., 2019). Thus, the inhibitory effects of FC on protein expression of GFAP and C3 and production of pro-inflammatory mediators in 2-CE-induced RAs could all be attributed to decreased RAs activity. It is reasonable to speculate that the inhibitory effects of FC pretreated RAs on the

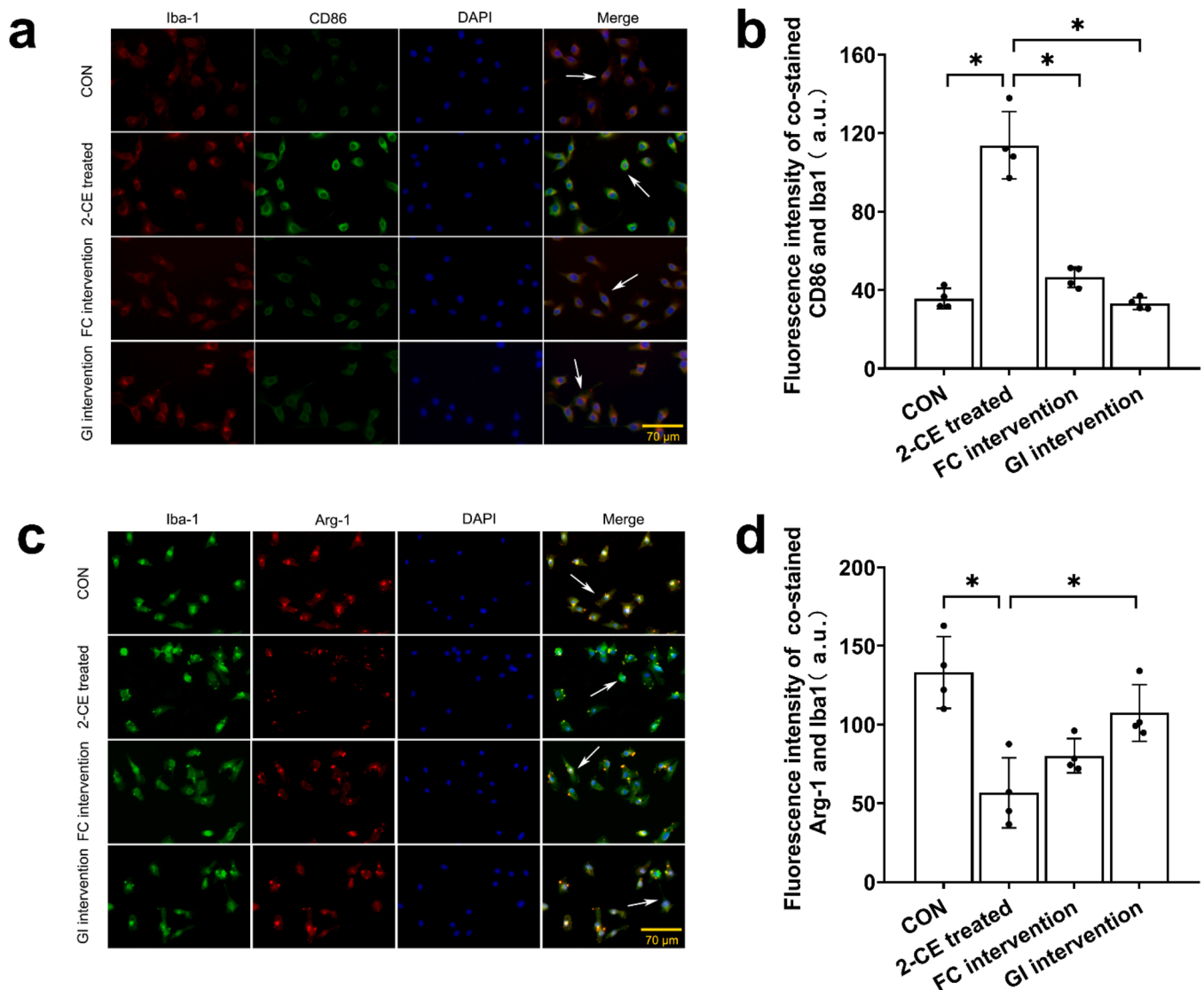


Fig. 5. Effects of FC and GI pretreatment on CD86 and Arg-1 protein expression in rat microglia via inhibition of 2-CE-induced RAs. (a) Immunostaining of microglial cells with Iba1 as well as pro-inflammatory-marker CD86 (200 ×, scale bar = 70 μm). (b) Quantification of co-stained CD86 and Iba1 fluorescence intensity (a.u.) in the corresponding images. (c) Immunostaining of microglial cells with the Iba1 as well as anti-inflammatory-marker Arg-1 (200 ×, scale bar = 70 μm). (d) Quantification of co-stained Arg-1 and Iba1 fluorescence intensity (a.u.) in the corresponding images. (n = 4; mean ± SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, * $p < 0.05$).

polarization of co-cultured microglia could be also associated with decreased RAs activity.

As a new anti-neuroinflammatory agent, GI, a pyridazine-3-carboxamide derivative is approved by China Food and Drug Administration to be used in clinical trials for Alzheimer's disease (AD) (Zhou et al., 2016). However, the effects of GI on RAs, and its application in treatment of 1,2-DCE poisoning has not been reported. Our study disclosed for the first time that GI treatment could abolish the pro-inflammatory effects of 2-CE induced RAs through suppressing of p38 MAPK/AP-1 and NF-κB signaling pathways. The effects of GI on 2-CE-induced RAs could also suppress the pro-inflammatory polarization and enhance anti-inflammatory polarization of co-cultured microglia.

Dia, an anthraquinone derivative and a nonsteroidal anti-inflammatory drug, has been reported to play a therapeutic role in osteoarthritis, diabetes, and kidney injury. Furthermore, the findings from recent studies revealed that Dia could attenuate brain edema, minimize hemorrhagic transformation, and improve neurological outcomes in an animal model of ischemic stroke (Chen et al., 2021).

However, the effects of Dia on RAs, and the role of Dia in treatment of 1, 2-DCE poisoning remains unknown. Our data revealed for the first time that Dia treatment could abolish the pro-inflammatory effects of 2-CE induced RAs through suppressing of p38 MAPK/NF-κB signaling pathway. The effects of Dia on 2-CE-induced RAs could also suppress the pro-inflammatory polarization and enhance anti-inflammatory polarization of co-cultured microglia. However, since Dia treatment alone could activate the AP-1 signaling pathway in astrocytes, Dia pretreatment failed to affect the AP-1 signaling pathway in 2-CE-induced RAs, thus it might be not the suitable candidate for therapeutic use in 1,2-DCE poisoning.

5. Conclusion

In conclusion, the present study demonstrated for the first time that supplementation of exogenous FC, GI or Dia might effectively ameliorate the pro-inflammatory effects of 2-CE-induced RAs through inhibition of p38 MAPK signaling pathways, and then further suppress the pro-inflammatory polarization of microglia induced by RAs during 2-CE

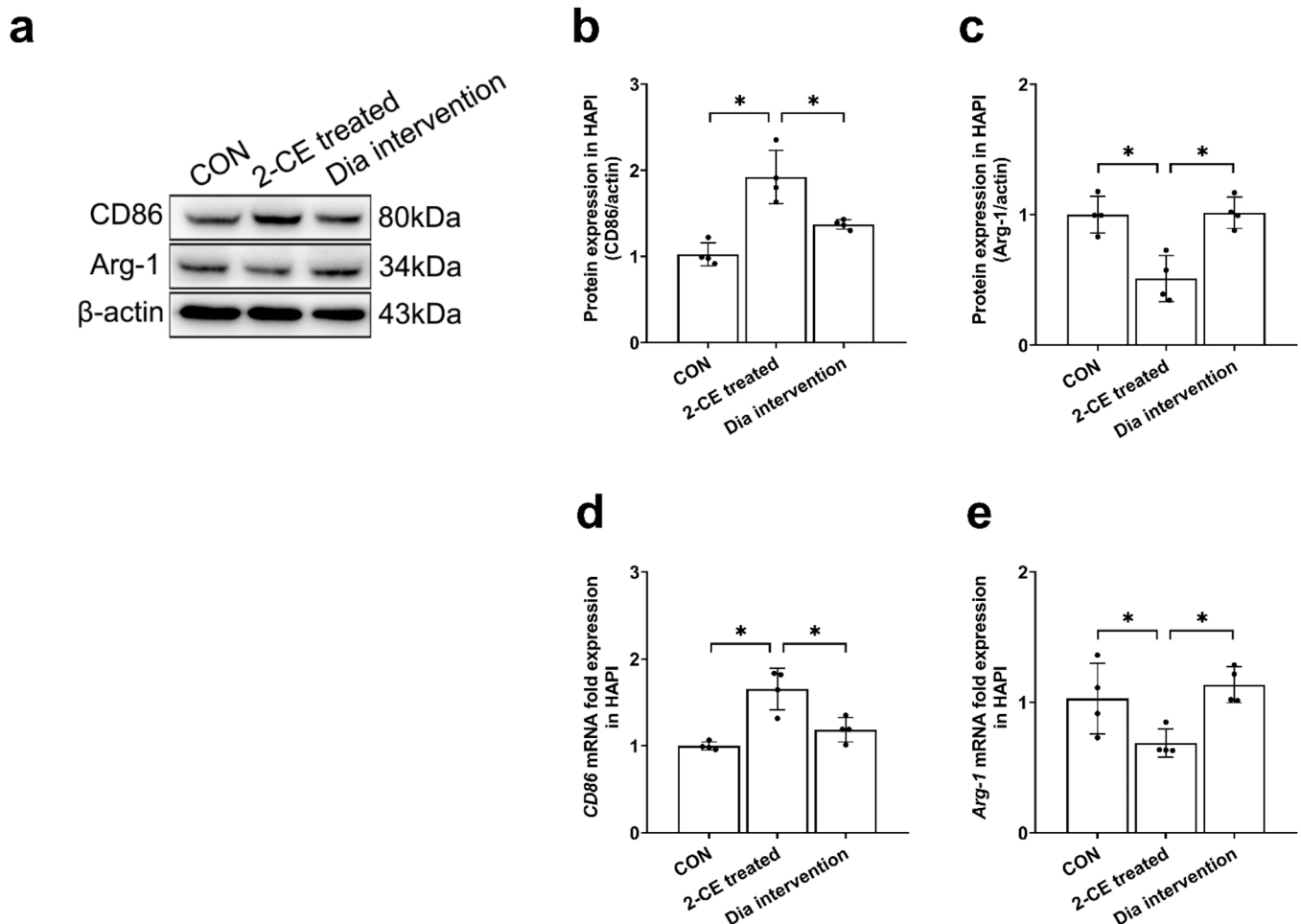


Fig. 6. Effects of Dia pretreatment on microglia polarization via inhibition of 2-CE-induced RAs. (a) Representative blots of CD86 and Arg-1 protein in HAPI microglial cells which were co-cultured with different conditioned astrocytes by transwell. (b, c) Quantification of CD86 and Arg-1 protein in HAPI microglial cells. (d, e) qPCR was used to assay the *CD86* and *Arg-1* mRNA levels in HAPI microglial cells. ($n = 4$; mean $\pm$ SD; normal distribution data; one-way ANOVA followed by SNK post hoc testing, * $p < 0.05$). Notes: HAPI microglial cells, highly aggressively proliferating immortalized microglial cells.

exposure or 1,2-DCE intoxication. Among these three tested agents, GI might be the most suitable candidate for preventing the neuroinflammation during the course of brain edema formation induced by subacute poisoning of 1,2-DCE.

Ethics approval and consent to participate

Animal experimental protocols were put forward in consistence with the guidelines of NIH on ethics and biosafety, which were approved by the Laboratory Animal Care and Use Committee of China Medical University (IACUC: No.2020108).

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Authors contribution

JYP and WT were in charge of the study design. WT conducted the experiments, data analysis, figure preparation, as well as manuscript drafting. YJH provided experimental technical support. ZFH and WGY reviewed the manuscript for important content. JYP edited and revised the manuscript. All authors have given approval to the final version of the manuscript.

CRediT authorship contribution statement

Yaping Jin and Tong Wang were in charge of the study design. Tong Wang conducted the experiments, data analysis, figure preparation, as well as manuscript drafting. Jinhan Yang provided experimental technical support. Fenghong Zhao and Gaoyang Wang reviewed the manuscript for important content. Yaping Jin edited and revised the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgements

Not applicable.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2023.115130](https://doi.org/10.1016/j.ecoenv.2023.115130).

References

- Almezgagi, M., et al., 2020. Diacerein: recent insight into pharmacological activities and molecular pathways. *Biomed. Pharm.* 131, 110594.
- Bharti, R., et al., 2017. Somatostatin receptor targeted liposomes with Diacerein inhibit IL-6 for breast cancer therapy. *Cancer Lett.* 388, 292–302.
- Chen, H., et al., 2021. Peroxynitrite activates NLRP3 inflammasome and contributes to hemorrhagic transformation and poor outcome in ischemic stroke with hyperglycemia. *Free Radic. Biol. Med.* 165, 171–183.
- Chen, S., et al., 2015. 1,2-Dichloroethane-induced toxic encephalopathy: a case series with morphological investigations. *J. Neurol. Sci.* 351, 36–40.
- Dhaya, I., et al., 2021. Magnetic resonance imaging under isoflurane anesthesia alters cortical cyclooxygenase-2 expression and glial cell morphology during sepsis-associated neurological dysfunction in rats. *Anim. Model Exp. Med.* 4, 249–260.
- Escartin, C., et al., 2021. Reactive astrocyte nomenclature, definitions, and future directions. *Nat. Neurosci.* 24, 312–325.
- Hawkins, B.T., Davis, T.P., 2005. The blood-brain barrier/neurovascular unit in health and disease. *Pharm. Rev.* 57, 173–185.
- Hoesel, B., Schmid, J.A., 2013. The complexity of NF-kappaB signaling in inflammation and cancer. *Mol. Cancer* 12, 86.
- Jin, X., et al., 2018. Involvement of CYP2E1 in the course of brain edema induced by subacute poisoning with 1,2-dichloroethane in mice. *Front Pharm.* 9, 1317.
- Jin, X., et al., 2019. Neuroinflammatory reactions in the brain of 1,2-DCE-intoxicated mice during brain edema. *Cells* 8.
- Lawrence, W.H., et al., 1971. Toxicity of ethylene chlorhydrin. I. Acute toxicity studies. *J. Pharm. Sci.* 60, 568–571.
- Li, R., et al., 2018. TSG-6 attenuates inflammation-induced brain injury via modulation of microglial polarization in SAH rats through the SOCS3/STAT3 pathway. *J. Neuroinflamm.* 15, 231.
- Li, T., et al., 2019. An update on reactive astrocytes in chronic pain. *J. Neuroinflamm.* 16, 140.
- Liddel, S., Barres, B., 2015. Snapshot: astrocytes in health and disease. *Cell* 162, 1170–1170 e1.
- Liddel, S.A., et al., 2017. Neurotoxic reactive astrocytes are induced by activated microglia. *Nature* 541, 481–487.
- Liddel, S.A., Barres, B.A., 2017. Reactive astrocytes: production, function, and therapeutic potential. *Immunity* 46, 957–967.
- Mocco, J., et al., 2006. Complement component C3 mediates inflammatory injury following focal cerebral ischemia. *Circ. Res.* 99, 209–217.
- Ni, Y., et al., 2017. TNFalpha alters occludin and cerebral endothelial permeability: role of p38MAPK. *PLoS One* 12, e0170346.
- Pekny, M., et al., 2019. Astrocyte activation and reactive gliosis—a new target in stroke? *Neurosci. Lett.* 689, 45–55.
- Rynkowski, M.A., et al., 2009. C3a receptor antagonist attenuates brain injury after intracerebral hemorrhage. *J. Cereb. Blood Flow. Metab.* 29, 98–107.
- Shang, X.L., et al., 2015. Fluorocitrate induced the alterations of memory-related proteins and tau hyperphosphorylation in SD rats. *Neurosci. Lett.* 584, 230–235.
- Subhramanyam, C.S., et al., 2019. Microglia-mediated neuroinflammation in neurodegenerative diseases. *Semin Cell Dev. Biol.* 94, 112–120.
- Wang, T., et al., 2017. Upregulation of matrix metalloproteinase-9 in primary cultured rat astrocytes induced by 2-chloroethanol Via MAPK signal pathways. *Front Cell Neurosci.* 11, 218.
- Wang, T., et al., 2018. Association of NF-kappaB and AP-1 with MMP-9 overexpression in 2-chloroethanol exposed rat astrocytes. *Cells* 7.
- Wang, T., et al., 2021. Reactive astrocytes induced by 2-chloroethanol modulate microglia polarization through IL-1beta, TNF-alpha, and iNOS upregulation. *Food Chem. Toxicol.* 157, 112550.
- Wang, Y., et al., 2019. Inhibition of activated astrocyte ameliorates lipopolysaccharide-induced depressive-like behaviors. *J. Affect Disord.* 242, 52–59.
- Wu, F., et al., 2018. Simvastatin alters M1/M2 polarization of murine BV2 microglia via Notch signaling. *J. Neuroimmunol.* 316, 56–64.
- Yang, J., et al., 2021. Roles of Crosstalk between Astrocytes and Microglia in Triggering Neuroinflammation and Brain Edema Formation in 1,2-Dichloroethane-Intoxicated Mice. *Cells* 10.
- Zhang, H.Y., et al., 2020. A1 astrocytes contribute to murine depression-like behavior and cognitive dysfunction, which can be alleviated by IL-10 or fluorocitrate treatment. *J. Neuroinflamm.* 17, 200.
- Zhang, X., et al., 2019. Inhibition of reactive astrocytes with fluorocitrate ameliorates learning and memory impairment through upregulating CRTCI and synaptophysin in ischemic stroke rats. *Cell Mol. Neurobiol.* 39, 1151–1163.
- Zhou, W., et al., 2016. Microglia-based phenotypic screening identifies a novel inhibitor of neuroinflammation effective in Alzheimer's disease models. *ACS Chem. Neurosci.* 7, 1499–1507.